

[Project Name TBD] — Product Requirements Document

Version 1.1 · 26 Sep 2026 · Status: final for submission · Working placeholder: "**the System**" (final name to be chosen by the team before the video is recorded)

This PRD is a living document: it is updated as the build progresses and becomes the record of the full implementation.

1. Context

- **Event:** Space4Innovation × Tama University, YOHAKU 2026. **Challenge 7: "What can we learn from mycelium? Fungi, AI and shared decisions in orbit"** (challenge leads: Diana Mastracci, Prof. Takashi Hikasa).
- **Deadline:** Sunday 27 September 2026, 14:00 CEST (17:30 IST). Internal target: submitted by 13:00 CEST, with one hour of buffer.
- **Required:** a solution in any form, plus a 2-minute video explaining it.

2. Problem statement

Orbit has become **orbital mixed traffic**. ESA counts about 11,600 functioning satellites among roughly 41,700 objects regularly tracked by surveillance networks (ESA, May 2025); most of the rest are debris, dead satellites and rocket bodies. Between June 2025 and May 2026, Starlink alone reported more than 355,000 collision-avoidance manoeuvres to the FCC, 207,152 of them in the last six months.

That figure partly reflects a policy choice: Starlink manoeuvres at a collision probability of $3e-7$, far stricter than the common $1e-4$ threshold. Every manoeuvre also makes everyone else's orbit predictions for that satellite stale. **One operator's safety choice becomes a cost for its neighbours**, which is exactly the kind of shared burden this project addresses.

There is **no universal right-of-way**, and coordination is still ad hoc:

- **September 2019:** ESA's Aeolus manoeuvred alone to avoid Starlink 44. SpaceX later said a bug in its on-call paging system meant ESA's follow-up messages were never seen.
- **December 2025:** SpaceX reported that a satellite deployed from a Chinese Kinetica-1 launch passed about 200 m from Starlink-6079, saying no coordination had been done. The launch provider, CAS Space, said it was looking into the claim. The facts remain disputed, which is itself the point: there was no shared record to check.

As AI takes over more of these decisions, key questions remain open:

- Who moves when both objects can? What changes when one is debris?
- Who pays the fuel cost of repeated avoidance? Should a satellite's purpose, or its contribution to the risk, matter?
- Where should AI sit, and when must a human step in?
- Who is responsible when an AI-shaped decision causes harm? Who decides what the network learns?
- **NARETU:** what does each actor owe the others, and those who come after?

3. Solution overview

The System is a coordination system inspired by mycorrhizal (fungal) networks.

Fungal network	The System
Hyphal tips sense and respond locally	Every satellite carries the same small trained AI policy , which senses risk and decides on its own
One connected network	Satellites broadcast intent to their neighbours over "hyphae" links (shared data plus a handshake)
A market with sanctions (Kiers et al. 2011)	A fairness ledger rewards those who yield and penalises free-riders
What nature lacks	Shared right-of-way rules , human escalation , responsibility allocation , governed learning and obligations to the future (NARETU)

Tagline: "Borrow the fungus's architecture, not its morality."

4. Goals and non-goals

Goals	Non-goals
G1: A working decentralised prototype, with a trained AI if time allows (rules-only fallback is acceptable)	Operational-grade collision probability or orbit determination
G2: An explicit answer to every challenge question	Integration with real operator systems
G3: An honest analogy map (holds / bends / breaks)	Certified safety or legal advice
G4: An answer to the NARETU obligations question	A native mobile app
G5: A compelling 2-minute video and a demo link that opens	

5. Personas

- **P1, Judges:** interested in mixed-traffic ethics, priority for vulnerable users, and governance. They need to grasp the idea in 2 minutes, then explore it.
- **P2, Satellite operator:** uses the dashboard. Sees alerts and AI decisions, and approves or overrides them.
- **P3, Regulator / incident board:** reviews the decision log and approves learning updates.

6. Features

6.1 Core demo features

ID	Feature	Description	Acceptance criteria
F1	Space Map	3D Earth showing about 3,000 real objects from a CelesTrak snapshot, coloured by type: grey = debris, blue = manoeuvrable, green = autonomous, gold = crewed. Time-lapse playback.	Loads in under 5 s; at least 30 fps; legend shown; credit "Data: CelesTrak"
F2	Danger Threads (hyphae)	Glowing arcs link objects on a close pass. They get thicker and brighter as risk rises and fade afterwards (a slime-mould-style reinforce-and-decay rule).	Thread lifecycle visible for every scripted event
F3	Onboard AI Brain	A trained, shared RL policy runs live in the browser for each satellite. A panel shows what the satellite sees, its action probabilities and the action it chose.	Browser output matches Python within 1e-4 on 100 test vectors

ID	Feature	Description	Acceptance criteria
F4	Who-Yields Rules	Right-of-way rules in the style of maritime COLREGs. Priority: debris and non-maneuvrable objects hold course, then crewed, then low-fuel, then science / public-good, then commercial. Tie-breaks: larger fuel margin yields, then ledger balance, then a deterministic ID. The object holding course must act by a deadline if the other side stays silent. A counterpart that does not answer the handshake is treated as non-maneuvrable, and its silence is recorded in the ledger. Priority class and fuel state are declared through the handshake and must match the operator's registered, state-verified class; a false declaration is a ledger penalty and evidence of fault (F14).	Deterministic: both satellites compute the same verdict from the shared handshake data only , never from private estimates
F5	Intent Handshake Feed	Chat-style log of PROPOSE / ACK / DO-NOT-MOVE / EXECUTED / ESCALATE messages	Visible for each event, with timestamps
F6	Fairness Ledger	Per operator: fuel spent, avoidance manoeuvres absorbed, risk contributed, debris legacy. Yielding earns credit; free-riders lose priority. Optional orbit-use fee pool, with a discount for emerging space nations.	Updates after each event; bar chart
F7	Human Escalation Ladder	Autonomy levels L0-L4. A human is called in when $P_c > 1e-3$, a crewed vehicle is involved, rules conflict, the counterpart is silent, the fuel cost is high, AI confidence is low, or the predictor raises an alarm. Modal offers Approve / Override / Stop. Überlingen rule: an agreed machine plan beats an ad-hoc override. Humans may override freely until both sides have sent EXECUTED-commit; after that, any change must re-enter the handshake and cannot be made unilaterally. Escalation triggers are hard rules outside the AI policy; the policy cannot suppress them.	Modal appears on trigger; the action is logged with who decided
F9	AI Explainer	A local open-weight LLM writes a plain-English reason for each decision. Explanations are pre-generated for the public site, with a template fallback.	Every scripted event has an explanation; the site works without any server
F10	2019 Aeolus Replay	Split screen: "Email world" (the real timeline) against "the System's world"	Runs in under 40 s; labelled "scripted from ESA's published timeline"
F11	Kessler Finale	"What if we fail?": one collision becomes thousands of fragments, and the threads multiply	Runs without the frame rate collapsing
F12	Results Panel	AI against 4 baselines on 1,000 test scenarios, plus learning curves and emergent-behaviour plots	Loaded from results.json

6.2 Deferred features (not built for this submission)

These stay in the design but are cut from the 27 September build. The Live API is reachable only inside a private network, so judges would never see it; the predictor does not answer any challenge question directly.

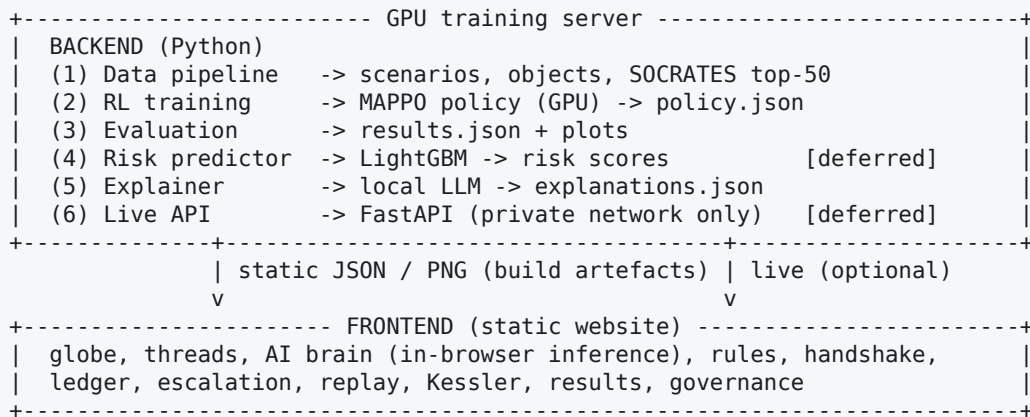
ID	Feature	Description
F8	Danger Predictor	LightGBM model trained on about 13,000 real ESA conjunction events, to predict early whether a warning will become dangerous, compared against ESA's "latest risk" baseline
F17	Live API mode	A small server on the GPU machine for live explanations, fresh scenarios and live benchmarks, with a LIVE / OFFLINE badge

6.3 Governance features (page sections and video)

ID	Feature	Description
F13	Where the Analogy Breaks	Table of where the fungal analogy holds, bends and breaks (Simard et al. 1997; Kiers et al. 2011; Karst et al. 2023)

ID	Feature	Description
F14	Responsibility Matrix	Failure modes against operator / AI developer / data provider / licensing state / international body. Evidence of fault: whether the rules were followed, and what the ledger shows. An operator that failed to share data or answer the handshake carries presumptive responsibility for the outcome.
F15	Learning Governance	The network logs every case. Changes to rules or models need approval from an independent "Orbital Incident Board", and every version is recorded.
F16	NARETU Obligations Charter	What each actor owes, to whom, and over what time horizon, including a cap on the debris legacy left to future generations. The NARETU framing is credited to the challenge brief and to Chief Titus Letaapo.

7. System architecture



Principle: static first. The website runs entirely from pre-built static files, so the link judges open always works.

Safety shield. The trained policy never acts alone. Every proposed action passes through the Who-Yields rules (F4) and the escalation triggers (F7) before it is executed. If the policy and the rules disagree, the rules win and the case is escalated.

7.1 Live API (deferred)

A small FastAPI server on the GPU machine. The website sends it requests and receives results.

Endpoint	Purpose	Uses
GET /health	Availability check; drives the LIVE / OFFLINE badge	—
POST /explain	Send a decision, optionally with a question; get a plain-English answer	Local LLM
POST /simulate	Send settings (number of satellites, debris, fuel levels); get fresh episodes run by the trained policy	RL policy (GPU)
POST /benchmark	Run N test scenarios now; get AI vs baseline metrics	RL environment (GPU)
POST /predict_risk	Send a conjunction's CDM history; get a danger score	LightGBM

Access: served over HTTPS inside a private network. Devices outside it, such as judges' browsers, fall back to OFFLINE mode automatically.

Guardrails:

- rate limit
- maximum scenario size

- an allow-list of site origins (CORS)
- the API runs only during demos

8. Backend (Python, GPU server)

Module	Purpose
sim/scenarios.py	Load the ESA Kelvins CSV (CC-BY-4.0) and bootstrap whole events into encounter geometry: miss vector, relative velocity, uncertainty schedule
sim/env_torch.py	Batched multi-agent GPU environment with 4,096 parallel encounters. Clohessy-Wiltshire dynamics projected onto the encounter plane. 24 steps of 10 minutes each.
sim/rules.py	Who-Yields rules engine (mirrored in JS)
sim/baselines.py	Four baselines: do nothing; both burn if $P_c > 1e-4$; lower ID yields; rules only
sim/train_mappo.py	PPO with a shared actor and a centralised critic. Curriculum: debris, then 2 satellites, then 2-6 satellites including crewed vehicles and third parties. 3 seeds; CSV logs.
sim/evaluate.py	Runs 1,000 held-out seeds and writes results.json plus plots: learning curves, baseline table, yield-vs-ledger curve, burn-timing histogram, along-track-only ($\pm T$) ablation
sim/risk_model.py	(Deferred) LightGBM on CDMs issued at least 2 days before TCA, compared with the latest-risk baseline
sim/explain.py	Batch LLM explanations, written to explanations.json
sim/export_policy.py	Exports weights and observation normalisation to policy.json, and 100 test vectors to testvec.json
sim/make_web_data.py	Builds the web data: a Celestrak subset to objects.json, the SOCRATES top 50 to socrates_top.json, and scripted episodes to episodes.json
sim/api.py	(Deferred) Live API (§7.1)
sim/tests/	Tests: CW burn produces the expected shift; P_c sanity; rule determinism; environment shapes

RL specification

- **Observation** (~24 floats):
 - time to TCA, miss vector, $\log P_c$, σ , relative velocity
 - own and neighbour fuel, ledger balance and class
 - the neighbour's declared intent
 - the offset already committed
 - the miss distance to any third party
- **Actions** (Discrete(7)):
 - hold
 - small opening burn, large opening burn, small closing burn
 - radial burn
 - request that the other object yields
 - escalate to a human
- **Reward**: $\beta \cdot \text{team} + (1-\beta) \cdot \text{self}$, made up of:
 - -100 for a collision
 - P_c shaping
 - -20 for creating a secondary conjunction
 - -fuel used
 - -fairness gap
 - -5 for escalating (applies only to discretionary escalation; hard triggers in F7 escalate regardless and carry no penalty, so the policy cannot learn to avoid human oversight)
 - -0.1 per yield request

- **Network and training:** MLP 64×64 with tanh; 4,096 parallel environments; learning rate 3e-4; $\gamma = 0.99$; $\lambda = 0.95$; clip 0.2.

9. Frontend (static website)

Tech: plain HTML with ES modules and no build step. Libraries from a CDN: globe.gl (three.js), satellite.js (SGP4) and Chart.js. Dark space theme plus a light mode; responsive, designed desktop-first.

Layout:

1. **Hero:** headline statistics and a "Launch demo" button.
2. **Mission Control:**
 - **Centre:** globe (F1) with threads (F2).
 - **Left panel:** event list and escalation status (F7).
 - **Right panel tabs:** AI Brain (F3), Handshake (F5), Explanation (F9), Ledger (F6).
 - **Top bar:**
 - time controls and a scenario picker
 - buttons: "Replay 2019" (F10), "What if we fail?" (F11)
 - a "Rules only vs AI" toggle
 - **Modal:** human decision (Approve / Override / Stop).
3. **Results** (F12).
4. **How it works:** diagram mapping mycelium to orbit.
5. **Governance sections:** where the analogy breaks (F13), responsibility (F14), learning governance (F15), NARETU charter (F16).
6. **Sources and credits.**

JS modules (in web/js/):

Module	Role
globe.js	Globe rendering
threads.js	Danger threads
env.js	CW dynamics port
policy.js	MLP forward pass and parity test
rules.js	Who-Yields rules
ledger.js	Fairness ledger
escalation.js	Human escalation
explainer.js	Explanations: pre-generated, then template
replay.js	2019 Aeolus replay
kessler.js	Kessler finale
results.js	Results panel
ui.js	Page wiring

Data (in web/data/): objects.json, socrates_top.json, episodes.json, policy.json, testvec.json, explanations.json, results.json.

Performance:

- at most 3,000 points
- positions propagated about once a second, not every frame
- objects drawn as points, not meshes

10. Repository

```
README.md · LICENSE (MIT) · .gitignore
docs/ PRD.md · governance.md · method.md · submission-answers.md
sim/  backend
web/  frontend
```

- The repository is private for now and will be made public if the submission requires it.
- The Live API URL is kept in a local, uncommitted config file. A `config.example.js` is committed instead.
- **Credits:**
 - ESA Kelvins Collision Avoidance dataset: CC-BY-4.0, Zenodo 10.5281/zenodo.4463683 (Uriot et al. 2022)
 - CelesTrak (T.S. Kelso)
 - `globe.gl` and `satellite.js` (MIT)
 - No GPL code is included.

11. Non-functional requirements

- The static site works fully without any server.
- Every Pc value is labelled **illustrative**, and the use of linearised dynamics is stated.
- The site runs at 30 fps or better on a laptop and first loads in under 5 s.

12. Success metrics

- **RL against baselines, over 1,000 scenarios:**
 - collisions no higher than the best rule baseline
 - less Δv per event
 - a lower burden Gini
 - escalations below 10%
- **Emergent behaviour:**
 - the probability of yielding rises with ledger imbalance
 - burns happen earlier and smaller, and are along-track
- **Governance:** every challenge question has a written answer that points to a specific feature.
- **Demo:** every MUST feature reachable from the public link, and the video no longer than 2:00.

13. Priorities

Tier	Scope
MUST	Video; submission answers; a demo link that opens; F1, F2, F4, F5, F6, F7, F10; F13–F16 as written sections
SHOULD	F3 (trained RL in the browser); F12 results; F9 explainer (static)
COULD	F11 Kessler finale; F15 as a diagram
WON'T (this submission)	F8 predictor; F17 Live API

14. Risks and fallbacks

Risk	Fallback
Multi-agent RL does not converge	Single-agent PPO against a rule-following partner, with the same policy on every satellite

Risk	Fallback
GPU or driver issue	Train on CPU instead
LLM unavailable	Template explanations
Time overrun	Stop at the end of Phase 2 if needed (§16). Otherwise drop in this order: F11, then F3 (show the rules-only system and state that RL is future work), then F12. Never cut F7, F10 or F13-F16.
RL finishes but does not beat the rules baseline	Report it honestly; the rules layer is the product and RL is an experiment on top

15. Video outline (2:00)

Time	Beat
0:00	Hook: 207,152 Starlink avoidance manoeuvres in six months, and the 2019 missed message
0:15	Mycelium lens , then the twist: it is a "market with sanctions"
0:35	Demo: the 2019 replay, the AI brain, the handshake and the ledger
1:10	Humans and AI: escalation, the Überlingen lesson, the learning board
1:30	Where the analogy breaks
1:45	NARETU charter , closing on "protect the vulnerable first"

16. Delivery phases

The build runs in six phases. Each phase has a clear exit criterion, and **a phase is not started until the previous one meets its exit criterion**, with one exception: RL training (Phase 3) runs in the background from Phase 1 onwards because it needs GPU time, not attention.

All times are CEST, with IST in brackets. Deadline: Sun 27 Sep, 14:00 (17:30).

Summary

Phase	Name	Tier	Target end (CEST)	Features
0	Foundations	MUST	Sat evening	Repo, data pipeline, scripted episodes
1	Core coordination	MUST	Sat night	F1, F2, F4, F5, F6, F7, safety shield
2	Story and governance	MUST	Sun 06:00 (09:30)	F10, F13-F16, submission answers
3	AI layer	SHOULD	Sun 08:00 (11:30), decision gate	F3, F12, F9
4	Polish	COULD	Sun 10:00 (13:30)	F11, F15 diagram
5	Ship	MUST	Sun 13:00 (16:30)	Deploy, video, one-pager, submission
6	After submission	Future	—	F8, F17, next steps from §18

Phase 0: Foundations

Goal: everything later phases need exists as data, so the frontend never waits on the backend.

Work	Modules
Repository skeleton, licence, .gitignore, config.example.js	§10
Load the ESA Kelvins CSV and build encounter geometry	sim/scenarios.py

Work	Modules
CelesTrak subset and SOCRATES top 50	sim/make_web_data.py → objects.json, socrates_top.json
Hand-script the demo events, including the 2019 Aeolus timeline	episodes.json
Batched GPU environment and unit tests	sim/env_torch.py, sim/tests/

Exit criterion: `pytest sim/tests` passes and all web data files load in the browser.

Phase 1: Core coordination

Goal: a rules-only version of the System that works end to end. This is the minimum credible product; everything after it is additive.

Work	Features
Globe with real objects, coloured by type	F1
Danger threads with reinforce-and-decay	F2
Who-Yields rules in Python and JS, including silent counterparts and verified class	F4, sim/rules.py, rules.js
Handshake feed	F5
Fairness ledger	F6
Escalation ladder, hard triggers and override window	F7
Safety shield wiring: every action passes through F4 and F7	\$7
Background: start MAPPO training with the curriculum	sim/train_mappo.py

Exit criterion: a scripted conflict plays out on the globe, both sides reach the same verdict, the handshake and ledger update, and a hard trigger opens the human modal.

Phase 2: Story and governance

Goal: the parts judges will weigh most heavily. These answer the challenge questions directly.

Work	Features
2019 Aeolus replay: "Email world" against "the System's world"	F10
Where the analogy breaks (holds / bends / breaks)	F13
Responsibility matrix	F14
Learning governance (written section)	F15
NARETU obligations charter	F16
Draft all text answers	docs/submission-answers.md

Exit criterion: every MUST feature works, and every challenge question has a written answer that points to a feature. **If time runs out, the project can be submitted from here.**

Phase 3: AI layer (decision gate)

Goal: replace rules-only behaviour with the trained policy, running behind the safety shield.

Work	Features
Evaluate on 1,000 held-out seeds against 4 baselines	sim/evaluate.py → results.json
Export the policy and check browser parity	sim/export_policy.py, policy.js, F3
Results panel	F12

Work	Features
Pre-generate explanations, with template fallback	sim/explain.py, F9

Decision gate at Sun 08:00 (11:30):

- The policy matches or beats the best rules baseline on collisions → ship F3 and F12.
- It does not → ship rules-only, show the RL results honestly as an experiment, and state it as future work (§14).

Exit criterion: "policy parity OK" in the console, or the gate decision recorded.

Phase 4: Polish

Goal: optional impact. Only started if Phases 0–3 are closed.

Work	Features
Kessler finale	F11
Learning-governance diagram	F15 (diagram)
Light mode, responsive checks, performance pass (§11)	§9

Exit criterion: code freeze at **Sun 10:00 (13:30)**, whatever state Phase 4 is in.

Phase 5: Ship

Time (CEST)	Work
10:00 (13:30)	Code freeze. Deploy to the public URL; test from a device that is not the build machine
10:30 (14:00)	Record the demo screen capture following §15
11:30 (15:00)	Voice-over and edit to no more than 2:00; export the governance one-pager PDF
12:30 (16:00)	Fill in the form fields (§18), including the AI-use disclosure
13:00 (16:30)	Submit. One hour kept as buffer

Exit criterion: submission confirmation received.

Phase 6: After submission

Not part of the 27 September deliverable. Listed so the design stays whole.

- F8 Danger Predictor and F17 Live API (§6.2).
- The "From Obligation to Action" next steps: an open handshake protocol spec, a ledger pilot on existing data-sharing platforms, input to priority-rule standards, and a university mixed-traffic testbed.
- Make the repository public if the organisers request it.

17. Verification

- `pytest sim/tests` passes, and the collision rate in evaluation is below 1% at each curriculum stage.
- The browser console shows "policy parity OK" (maximum error below $1e-4$).
- Manual walkthrough: every MUST feature is visible from the public link, opened on a device that is not the build machine.

18. Submission (YOHAKU 2026 final form)

Field	Deliverable
Team name, solution title, members, location, organisation, bios	To be decided and filled in by the team
2-minute video (required)	Screen recording of the demo with the team's own voice-over
Final work link (required)	Demo website URL and a governance one-pager (PDF)
Code repository	This repository (MIT)
Relationships & Consequences	Answer draws on F4, F6, F7
Knowledge, Authority & Reciprocity	Answer draws on F13 and the credits
Orbital Stewardship & Justice	Answer draws on F6, F11, F16
Responsible AI & Accountability	Answer draws on F3, F7, F14, F15
From Obligation to Action	Next steps: open handshake protocol spec; ledger pilot on existing data-sharing platforms; input to priority-rule standards; university mixed-traffic testbed
AI use	An honest disclosure of the AI tools used during the build and of the AI components in the solution

Full drafts of the text answers are in docs/submission-answers.md.

19. Revision history

Version	Date	Changes
0.3	26 Sep 2026	Approved for build
1.0	26 Sep 2026	Corrected orbital statistics and the December 2025 incident to match published sources; added a safety shield so rules and escalation triggers sit outside the AI policy; closed the escalation-penalty loophole in the reward; defined silent and non-cooperative counterparts; added verification of declared priority class; clarified when a human override is allowed; deferred F8 and F17; fixed the baseline count; added a build schedule
1.1	26 Sep 2026	Replaced the build schedule with six delivery phases, each with scope, exit criteria and a decision gate for the AI layer